MTRC

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References Book:

1. **COMPUTER NETWORK,ANDREW S. TANENBAUM 5TH EDITION PEARSON PUBLICTION**
2. **COMPUTER NETWORKING JAMES F KUROSE PEARSON PUBLICATION**
3. **COMPUTER NETWORKING BEHROUZ A FOROUZAN TMH PUBLICATION**

17/10/2024

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UNIT 5

APPLICATION LAYER

Introduction:

The Application Layer is topmost layer in the Open System Interconnection (OSI) model. This layer provides several ways for manipulating the data (information) which actually enables any type of user to access network with ease. This layer also makes a request to its bottom layer, which is presentation layer for receiving various types of information from it. The Application Layer interface directly interacts with application and provides common web application services. This layer is basically highest level of open system, which provides services directly for application process.
Present Layer=> Application Layer

Presentation Layer

Session Layer

Transport Layer

Network Layer

Data Layer

Physical Layer

Functions of Application Layer:

The Application Layer, as discussed above, being topmost layer in OSI model, performs several kinds of functions which are requirement in any kind of application or communication process.
Following are list of functions which are performed by Application Layer of OSI Model –
Data from User <=> Application layer <=> Data from Presentation Layer

- Application Layer provides a facility by which users can forward several emails and it also provides a storage facility.
- This layer allows users to access, retrieve and manage files in a remote computer.
- It allows users to log on as a remote host.
- This layer provides access to global information about various services.
- This layer provides services which include: e-mail, transferring files, distributing results to the user, directory services, network resources and so on.
- It provides protocols that allow software to send and receive information and present meaningful data to users.
- It handles issues such as network transparency, resource allocation and so on.
- This layer serves as a window for users and application processes to access network services.
- Application Layer is basically not a function, but it performs application layer functions.
- The application layer is actually an abstraction layer that specifies the shared protocols and interface methods used by hosts in a communication network.
- Application Layer helps us to identify communication partners, and synchronizing communication.
- This layer allows users to interact with other software applications.
- In this layer, data is in visual form, which makes users truly understand data rather than remembering or visualize the data in the binary format (0's or 1's).

- This application layer basically interacts with Operating System (OS) and thus further preserves the data in a suitable manner.
- This layer also receives and preserves data from it's previous layer, which is Presentation Layer (which carries in itself the syntax and semantics of the information transmitted).
- The protocols which are used in this application layer depend upon what information users wish to send or receive.
- This application layer, in general, performs host initialization followed by remote login to hosts.

Working of Application Layer in the OSI model:

In the OSI model, this application layer is narrower in scope. The application layer in the OSI model generally acts only like the interface which is responsible for communicating with host-based and user-facing applications. This is in contrast with TCP/IP protocol, wherein the layers below the application layer, which is Session Layer and Presentation layer, are clubbed together and form a simple single layer which is responsible for performing the functions, which includes controlling the dialogues between computers, establishing as well as maintaining as well as ending a particular session, providing data compression and data encryption and so on.
At first, client sends a command to server and when server receives that command, it allocates port number to client. Thereafter, the client sends an initiation connection request to server and when server receives request, it gives acknowledgement (ACK) to client through client has successfully established a connection with the server and, therefore, now client has access to server through which it may either ask server to send any types of files or other documents or it may upload some files or documents on server itself.

Features provided by Application Layer Protocols:

To ensure smooth communication, application layer protocols are implemented the same on source host and destination host.

The following are some of the features which are provided by Application layer protocols-
- The Application Layer protocol defines process for both parties which are involved in communication.
- These protocols define the type of message being sent or received from any side (either source host or destination host).
- These protocols also define basic syntax of the message being forwarded or retrieved.
- These protocols define the way to send a message and the expected response.
- These protocols also define interaction with the next level.

Application Layer Protocols: The application layer provides several protocols which allow any software to easily send and receive information and present meaningful data to its users. The following are some of the protocols which are provided by the application layer.
- **TELNET:** Telnet stands for Telecommunications Network. This protocol is used for managing files over the Internet. It allows the Telnet clients to access the resources of Telnet server. Telnet uses port number 23.
- **DNS:** DNS stands for Domain Name System. The DNS service translates the domain name (selected by user) into the corresponding IP address. For example- If you choose the domain name as www.abcd.com, then DNS must translate it as 192.36.20.8 (random IP address written just for understanding purposes). DNS protocol uses the port number 53.
- **DHCP:** DHCP stands for Dynamic Host Configuration Protocol. It provides IP addresses to hosts. Whenever a host tries to register for an IP address with the DHCP server, DHCP

server provides lots of information to the corresponding host. DHCP uses port numbers 67 and 68.

- **FTP:** FTP stands for File Transfer Protocol. This protocol helps to transfer different files from one device to another. FTP promotes sharing of files via remote computer devices with reliable, efficient data transfer. FTP uses port number 20 for data access and port number 21 for data control.
- **SMTP:** SMTP stands for Simple Mail Transfer Protocol. It is used to transfer electronic mail from one user to another user. SMTP is used by end users to send emails with ease. SMTP uses port numbers 25 and 587.
- **HTTP:** HTTP stands for Hyper Text Transfer Protocol. It is the foundation of the World Wide Web (WWW). HTTP works on the client server model. This protocol is used for transmitting hypermedia documents like HTML. This protocol was designed particularly for the communications between the web browsers and web servers, but this protocol can also be used for several other purposes. HTTP is a stateless protocol (network protocol in which a client sends requests to server and server responses back as per the given state), which means the server is not responsible for maintaining the previous client's requests. HTTP uses port number 80.
- **NFS:** NFS stands for Network File System. This protocol allows remote hosts to mount files over a network and interact with those file systems as though they are mounted locally. NFS uses the port number 2049.
- **SNMP:** SNMP stands for Simple Network Management Protocol. This protocol gathers data by polling the devices from the network to the management station at fixed or random intervals, requiring them to disclose certain information. SNMP uses port numbers 161 (TCP) and 162 (UDP)

DOMAIN NAME SYSTEM (DNS) IN APPLICATION LAYER

Domain Name System (DNS) is a hostname for **IP address** translation service. DNS is a distributed database implemented in a hierarchy of name servers. It is an application layer protocol for message exchange between clients and servers. It is required for the functioning of the Internet.

What is the Need of DNS?

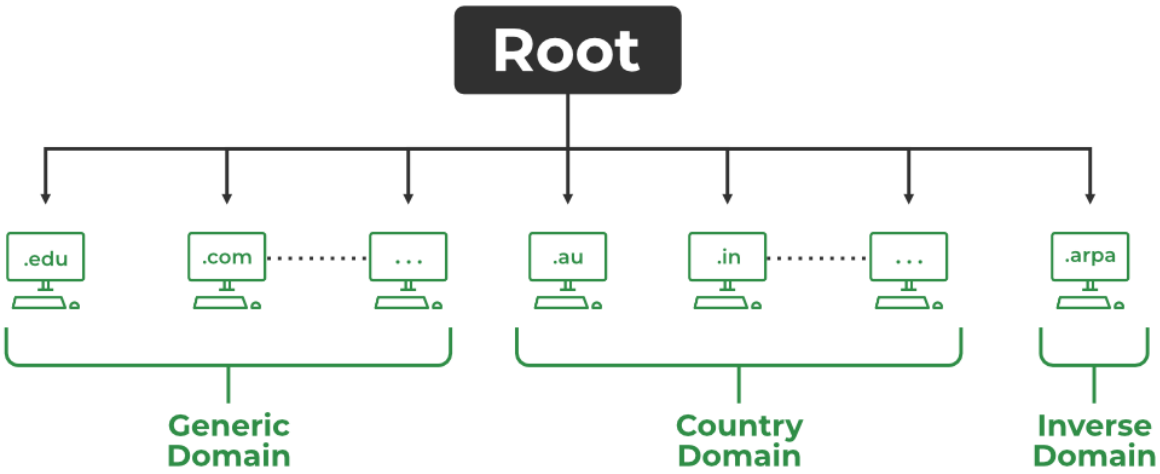
Every host is identified by the IP address but remembering numbers is very difficult for people also the IP addresses are not static therefore a mapping is required to change the domain name to the IP address. So DNS is used to convert the domain name of the websites to their numerical IP address.

Types of Domain

There are various kinds of domain:

1. **Generic domains:** .com (commercial), .edu (educational), .mil(military), .org(nonprofit organization), .net(similar to commercial) all these are generic domains.
2. **Country domain:** .in (India) .us .uk
3. **Inverse domain:** if we want to know what is the domain name of the website. Ip to domain name mapping. So DNS can provide both the mapping for example to find the IP addresses of geeksforgeeks.org then we have to type

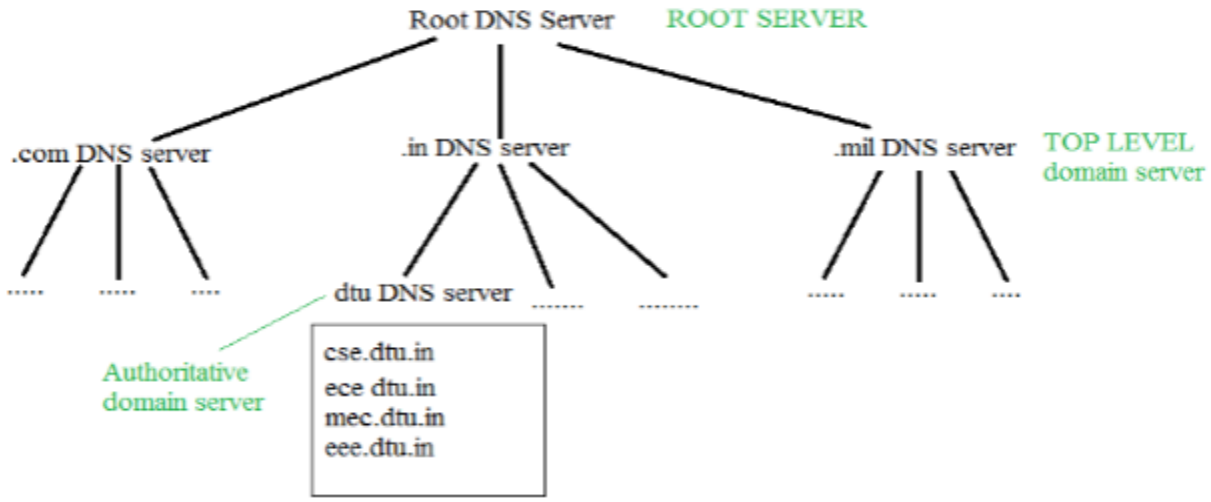
nslookup www.geeksforgeeks.org



Types of DNS

Organization of Domain

It is very difficult to find out the IP address associated with a website because there are millions of websites and with all those websites we should be able to generate the IP address immediately, there should not be a lot of delays for that to happen organization of the database is very important.



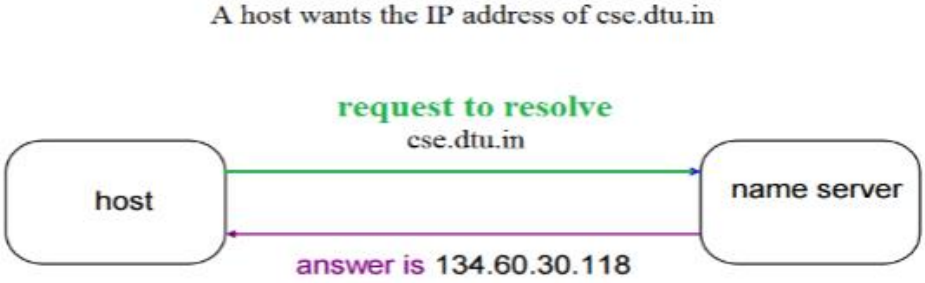
Root DNS Server

- **DNS record:** Domain name, IP addresses what is the validity? What is the time to live? and all the information related to that domain name. These records are stored in a tree-like structure.
- **Namespace:** Set of possible names, flat or hierarchical. The naming system maintains a collection of bindings of names to values – given a name; a resolution mechanism returns the corresponding value.
- **Name server:** It is an implementation of the resolution mechanism.

DNS = Name service in Internet – A zone is an administrative unit, and a domain is a sub tree.

Name-to-Address Resolution

The host requests the DNS name server to resolve the domain name. And the name server returns the IP address corresponding to that domain name to the host so that the host can future connect to that IP address.

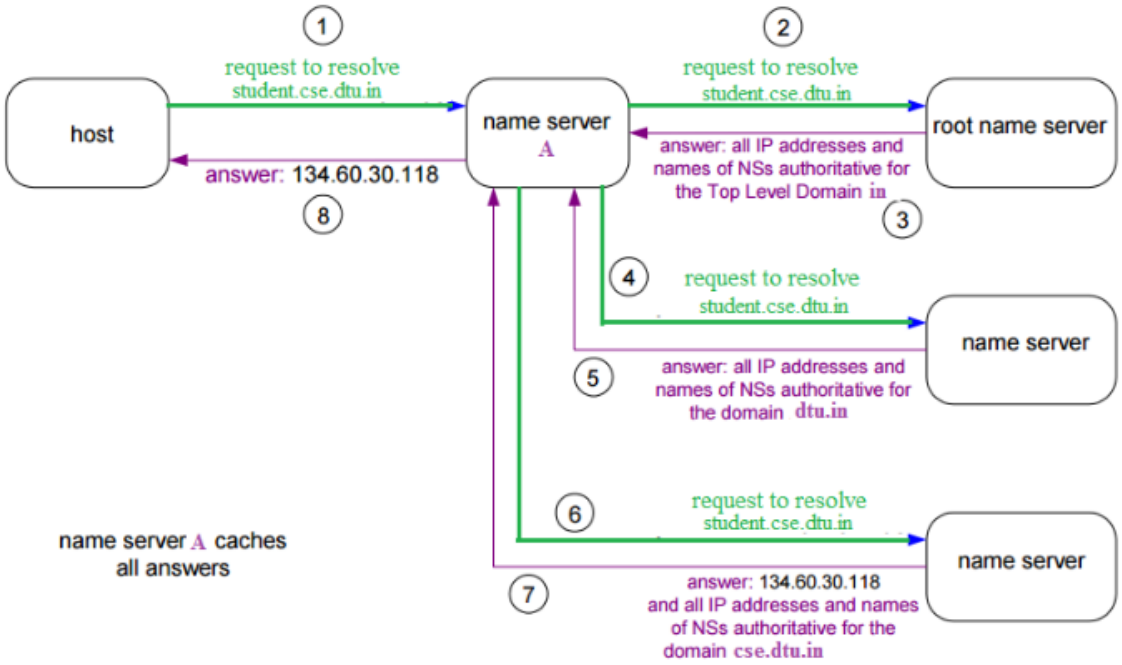


Name-to-Address Resolution

- **Hierarchy of Name Servers Root name servers:** It is contacted by name servers that can not resolve the name. It contacts the authoritative name server if name mapping is not known. It then gets the mapping and returns the IP address to the host.
- **Top-level domain (TLD) server:** It is responsible for com, org, edu, etc, and all top-level country domains like uk, fr, ca, in, etc. They have info about authoritative domain servers and know the names and IP addresses of each authoritative name server for the second-level domains.
- **Authoritative name servers** are the organization's DNS servers, providing authoritative hostnames to IP mapping for organization servers. It can be maintained by an organization or service provider. In order to reach cse.dtu.in we have to ask the root DNS server, then it will point out to the top-level domain server and then to the authoritative domain name server which actually contains the IP address. So the authoritative domain server will return the associative IP address.

Domain Name Server

The client machine sends a request to the local name server, which, if the root does not find the address in its database, sends a request to the root name server, which in turn, will route the query to a top-level domain (TLD) or authoritative name server. The root name server can also contain some hostname to IP address mappings. The Top-level domain (TLD) server always knows who the authoritative name server is. So finally the IP address is returned to the local name server which in turn returns the IP address to the host.

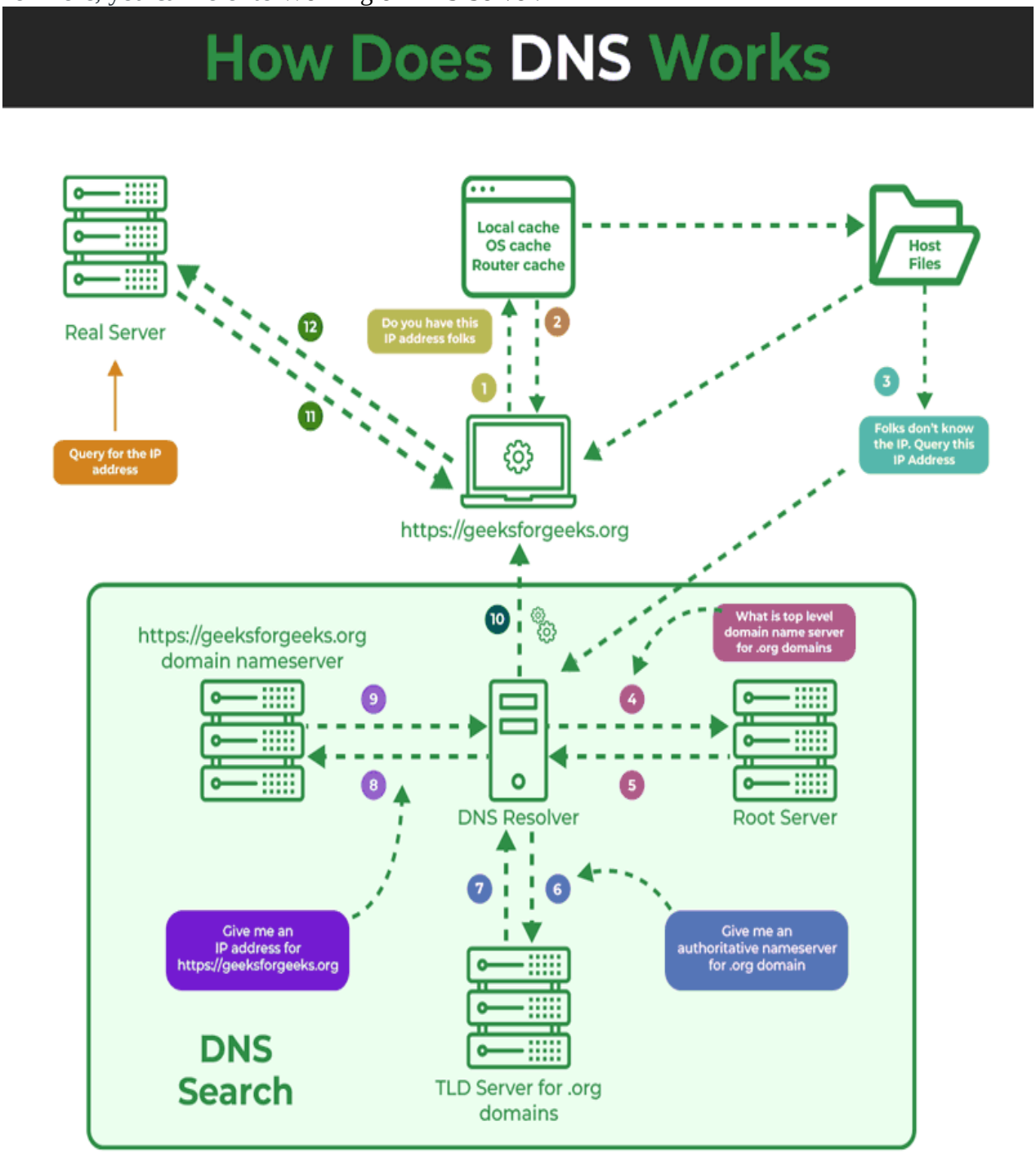


Domain Name Server

How Does DNS Work?

The working of DNS starts with converting a hostname into an IP Address. A domain name serves as a distinctive identification for a website. It is used in place of an IP address to make

it simpler for consumers to visit websites. Domain Name System works by executing the database whose work is to store the name of hosts which are available on the Internet. The top-level domain server stores address information for top-level domains such as .com and .net, .org, and so on. If the Client sends the request, then the DNS resolver sends a request to DNS Server to fetch the IP Address. In case, when it does not contain that particular IP Address with a hostname, it forwards the request to another DNS Server. When IP Address has arrived at the resolver, it completes the request over Internet Protocol.
For more, you can refer to Working of DNS Server.



ELECTRONIC MAIL

Introduction:

Electronic mail, commonly known as email, is a method of exchanging messages over the internet. Here are the basics of email:

1. An email address: This is a unique identifier for each user, typically in the format of name@domain.com.
2. An email client: This is a software program used to send, receive and manage emails, such as Gmail, Outlook, or Apple Mail.
3. An email server: This is a computer system responsible for storing and forwarding emails to their intended recipients.

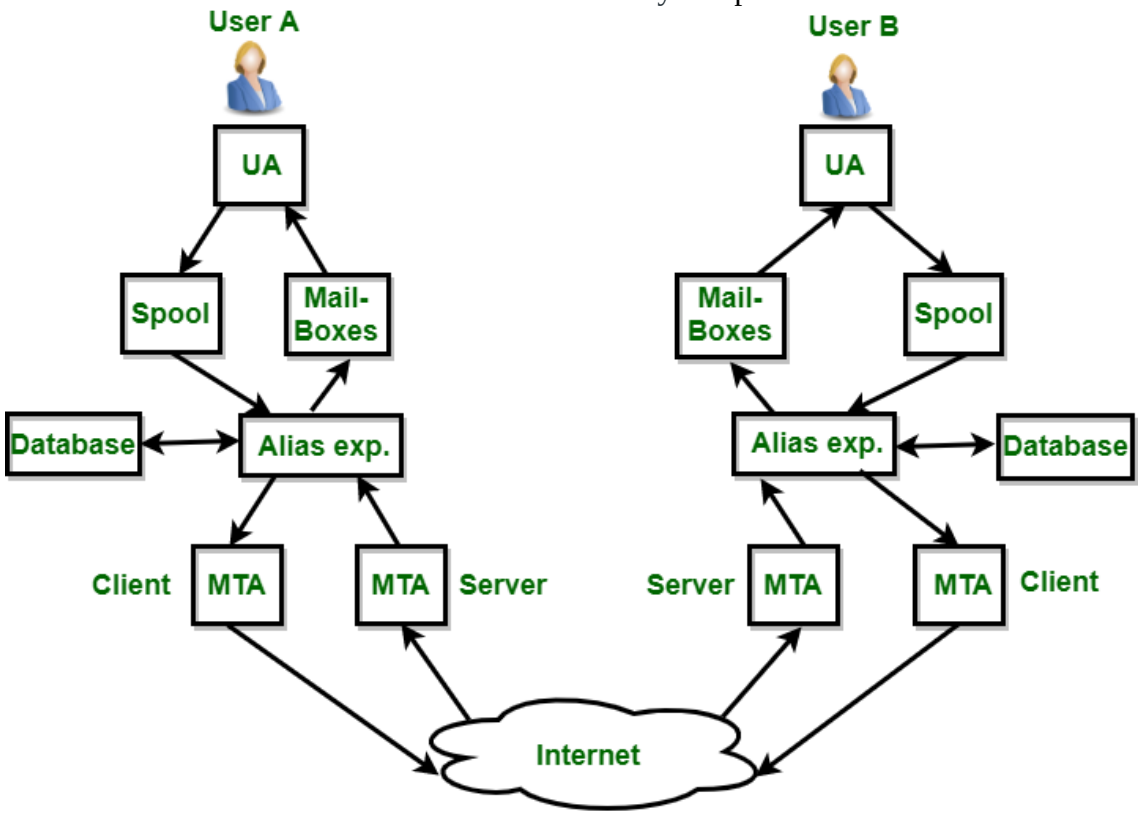
To send an email:

1. Compose a new message in your email client.
2. Enter the recipient's email address in the "To" field.
3. Add a subject line to summarize the content of the message.
4. Write the body of the message.
5. Attach any relevant files if needed.
6. Click "Send" to deliver the message to the recipient's email server.
7. Emails can also include features such as cc (carbon copy) and bcc (blind carbon copy) to send copies of the message to multiple recipients, and reply, reply all, and forward options to manage the conversation.

Electronic Mail (e-mail) is one of most widely used services of Internet. This service allows an Internet user to send a **message in formatted manner (mail)** to the other Internet user in any part of world. Message in mail not only contain text, but it also contains images, audio and videos data. The person who is sending mail is called **sender** and person who receives mail is called **recipient**. It is just like postal mail service. **Components of E-Mail System :** The basic components of an email system are : User Agent (UA), Message Transfer Agent (MTA), Mail Box, and Spool file. These are explained as following below.

1. **User Agent (UA) :** The UA is normally a program which is used to send and receive mail. Sometimes, it is called as mail reader. It accepts variety of commands for composing, receiving and replying to messages as well as for manipulation of the mailboxes.
2. **Message Transfer Agent (MTA) :** MTA is actually responsible for transfer of mail from one system to another. To send a mail, a system must have client MTA and system MTA. It transfer mail to mailboxes of recipients if they are connected in the same machine. It delivers mail to peer MTA if destination mailbox is in another machine. The delivery from

one MTA to another MTA is done by Simple Mail Transfer Protocol.



3. **Mailbox :** It is a file on local hard drive to collect mails. Delivered mails are present in this file. The user can read it delete it according to his/her requirement. To use e-mail system each user must have a mailbox . Access to mailbox is only to owner of mailbox.
4. **Spool file :** This file contains mails that are to be sent. User agent appends outgoing mails in this file using SMTP. MTA extracts pending mail from spool file for their delivery. E-mail allows one name, an **alias**, to represent several different e-mail addresses. It is known as **mailing list**, Whenever user have to sent a message, system checks recipient's name against alias database. If mailing list is present for defined alias, separate messages, one for each entry in the list, must be prepared and handed to MTA. If for defined alias, there is no such mailing list is present, name itself becomes naming address and a single message is delivered to mail transfer entity.

Services provided by E-mail system :

- **Composition –** The composition refer to process that creates messages and answers. For composition any kind of text editor can be used.
- **Transfer –** Transfer means sending procedure of mail i.e. from the sender to recipient.
- **Reporting –** Reporting refers to confirmation for delivery of mail. It help user to check whether their mail is delivered, lost or rejected.
- **Displaying –** It refers to present mail in form that is understand by the user.
- **Disposition –** This step concern with recipient that what will recipient do after receiving mail i.e save mail, delete before reading or delete after reading.

Advantages Or Disadvantages:

Advantages of email:

1. Convenient and fast communication with individuals or groups globally.
2. Easy to store and search for past messages.

3. Ability to send and receive attachments such as documents, images, and videos.
4. Cost-effective compared to traditional mail and fax.
5. Available 24/7.

Disadvantages of email:

1. Risk of spam and phishing attacks.
2. Overwhelming amount of emails can lead to information overload.
3. Can lead to decreased face-to-face communication and loss of personal touch.
4. Potential for miscommunication due to lack of tone and body language in written messages.
5. Technical issues, such as server outages, can disrupt email service.
6. It is important to use email responsibly and effectively, for example, by keeping the subject line clear and concise, using proper etiquette, and protecting against security threats.

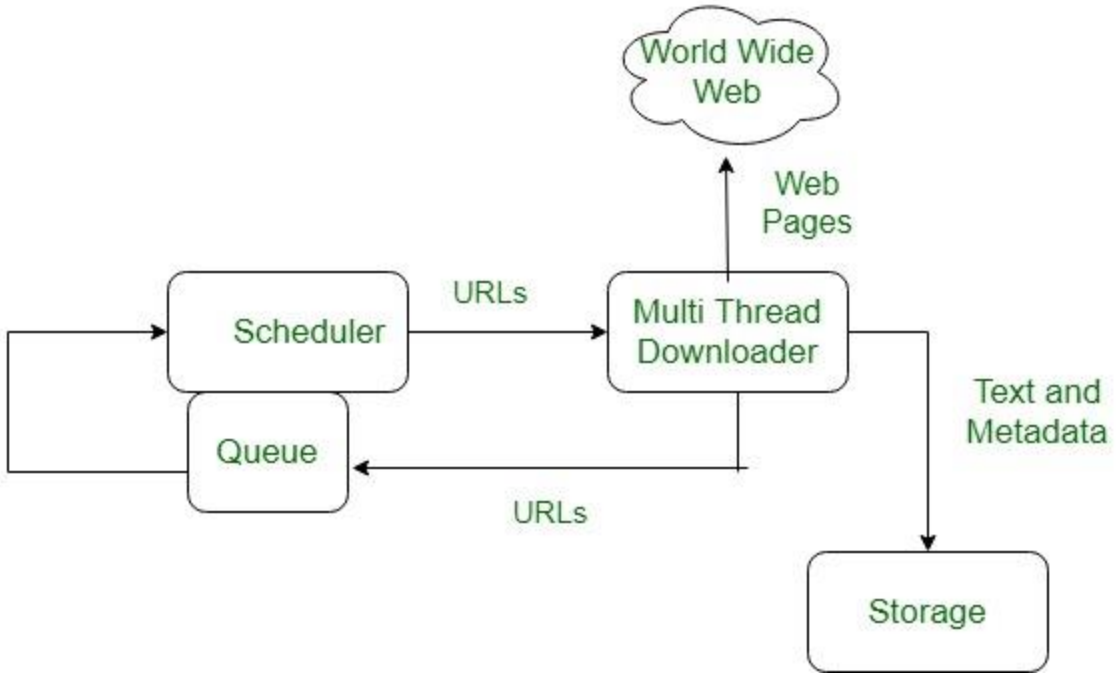
WORLD WIDE WEB (WWW)

The **World Wide Web** is abbreviated as WWW and is commonly known as the web. The WWW was initiated by CERN (European library for Nuclear Research) in 1989.
WWW can be defined as the collection of different websites around the world, containing different information shared via local servers(or computers).

History:
It is a project created, by Timothy Berner Lee in 1989, for researchers to work together effectively at CERN. is an organization, named the World Wide Web Consortium (W3C), which was developed for further development of the web. This organization is directed by Tim Berner's Lee, aka the father of the web.

System Architecture:

From the user's point of view, the web consists of a vast, worldwide connection of documents or web pages. Each page may contain links to other pages anywhere in the world. The pages can be retrieved and viewed by using browsers of which internet explorer, Netscape Navigator, Google Chrome, etc are the popular ones. The browser fetches the page requested interprets the text and formatting commands on it, and displays the page, properly formatted, on the screen.
The basic model of how the web works are shown in the figure below. Here the browser is displaying a web page on the client machine. When the user clicks on a line of text that is linked to a page on the abd.com server, the browser follows the hyperlink by sending a message to the abd.com server asking it for the page.



Here the browser displays a web page on the client machine when the user clicks on a line of text that is linked to a page on abd.com, the browser follows the hyperlink by sending a message to the abd.com server asking for the page.

Working of WWW:

The World Wide Web is based on several different technologies: Web browsers, Hypertext Markup Language (HTML) and Hypertext Transfer Protocol (HTTP).
A Web browser is used to access web pages. Web browsers can be defined as programs which display text, data, pictures, animation and video on the Internet. Hyperlinked resources on the World Wide Web can be accessed using software interfaces provided by Web browsers. Initially, Web browsers were used only for surfing the Web but now they have become more universal. Web browsers can be used for several tasks including conducting searches, mailing, transferring files, and much more. Some of the commonly used browsers are Internet Explorer, Opera Mini, and Google Chrome.

Features of WWW:

- HyperText Information System
- Cross-Platform
- Distributed
- Open Standards and Open Source
- Uses Web Browsers to provide a single interface for many services
- Dynamic, Interactive and Evolving.
- "Web 2.0"

Components of the Web: There are 3 components of the web:

1. **Uniform Resource Locator (URL):** serves as a system for resources on the web.
2. **HyperText Transfer Protocol (HTTP):** specifies communication of browser and server.
3. **Hyper Text Markup Language (HTML):** defines the structure, organisation and content of a webpage.

HTTP

HTTP stands for Hypertext Transfer Protocol. Tim Berner invents it. Hypertext is the type of text which is specially coded with the help of some standard coding language called Hypertext Mark-up Language (HTML). **HTTP/2** is the successor version of HTTP, which was published on May 2015. HTTP/3 is the latest version of HTTP, which is published in 2022.
The protocol used to transfer hypertext between two computers is known as Hypertext Transfer Protocol.

HTTP provides a standard between a web browser and a web server to establish communication. It is a set of rules for transferring data from one computer to another. Data such as text, images, and other multimedia files are shared on the World Wide Web. Whenever a web user opens their web browser, the user indirectly uses HTTP. It is an application protocol that is used for distributed, collaborative, hypermedia information systems.

Working of HTTP

First of all, whenever we want to open any website then first open a web browser after that we will type the URL of that website (e.g., www.facebook.com). This URL is now sent to Domain Name Server (DNS). Then DNS first check records for this URL in their database, then DNS will return the IP address to the web browser corresponding to this URL. Now the browser is able to send requests to the actual server.
After the server sends data to the client, the connection will be closed. If we want something else from the server we should have to re-establish the connection between the client and the server.

HTTP Connection

HTTP Request

HTTP request is simply termed as the information or data that is needed by Internet browsers for loading a website. This is simply known as HTTP Request.

There is some common information that is generally present in all HTTP requests. These are mentioned below.

- HTTP Version
- URL
- HTTP Method
- HTTP Request Headers
- HTTP Body

HTTP Request Headers

HTTP Request Headers generally store information in the form of key-value and must be present in each HTTP Request. The use of this Request Header is to provide core information about the client's information, etc.

HTTP Request Body

HTTP Request Body simply contains the information that has to be transferred. HTTP Request has the information or data to be sent to these browsers.

HTTP Method

HTTP Methods are simply HTTP Verb. In spite of being present so many HTTP Methods, the most common HTTP Methods are HTTP GET and HTTP POST. These two are generally used in HTTP cases. In HTTP GET, the information is received in the form of a website.

For more, refer to the Difference between HTTP GET and HTTP POST.

HTTP Response

HTTP Response is simply the answer to what a Server gets when the request is raised. There are various things contained in HTTP Response, some of them are listed below.

- HTTP Status Code
- HTTP Headers
- HTTP Body

HTTP Response

HTTP Response Headers

HTTP Response headers are simply like an HTTP Request where it has that work to send some important files and data to the HTTP Response Body.

HTTP Response Body

HTTP Responses are the responses that are received successfully upon the request. Generally, it comes under the requests generated by the web. In most cases, the request is of transferring the HTML data into a webpage.

HTTP Status Code

HTTP Status Codes are the 3-Digit codes that tell the message or simply tell us about the HTTP Request whether it has been completed or not. There are simply 5 types of status codes.

- Informational
- Successful
- Re-directional
- Client-Error
- Server-Error

History of HTTP

Tim Berners Lee and his team at CERN get credit for inventing original HTTP and associated technologies.

- **HTTP version 0.9:** This was the first version of HTTP which was introduced in 1991.
- **HTTP version 1.0:** In 1996, RFC 1945 (Request For Comments) was introduced in HTTP version 1.0.
- **HTTP version 1.1:** In January 1997, RFC 2068 was introduced in HTTP version 1.1. Improvements and updates to the HTTP version 1.1 standard were released under RFC 2616 in June 1999.

- **HTTP version 2.0:** The HTTP version 2.0 specification was published as RFC 7540 on May 14, 2015.
- **HTTP version 3.0:** HTTP version 3.0 is based on the previous RFC draft. It is renamed as Hyper-Text Transfer Protocol QUIC which is a transport layer network protocol developed by Google.

Characteristics of HTTP

HTTP is IP based communication protocol that is used to deliver data from server to client or vice-versa.

- The server processes a request, which is raised by the client, and also server and client know each other only during the current bid and response period.
- Any type of content can be exchanged as long as the server and client are compatible with it.
- Once data is exchanged, servers and clients are no longer connected.
- It is a request and response protocol based on client and server requirements.
- It is a connection-less protocol because after the connection is closed, the server does not remember anything about the client and the client does not remember anything about the server.
- It is a stateless protocol because both client and server do not expect anything from each other but they are still able to communicate.

Advantages of HTTP

- Memory usage and CPU usage are low because of fewer simultaneous connections.
- Since there are few TCP connections hence network congestion is less.
- Since handshaking is done at the initial connection stage, then latency is reduced because there is no further need for handshaking for subsequent requests.
- The error can be reported without closing the connection.
- HTTP allows HTTP pipe-lining of requests or responses.

Disadvantages of HTTP

- HTTP requires high power to establish communication and transfer data.
- HTTP is less secure because it does not use any encryption method like HTTPS and use TLS to encrypt regular HTTP requests and response.
- HTTP is not optimized for cellular phones and it is too gabby.
- HTTP does not offer a genuine exchange of data because it is less secure.
- The client does not close the connection until it receives complete data from the server; hence, the server needs to wait for data completion and cannot be available for other clients during this time

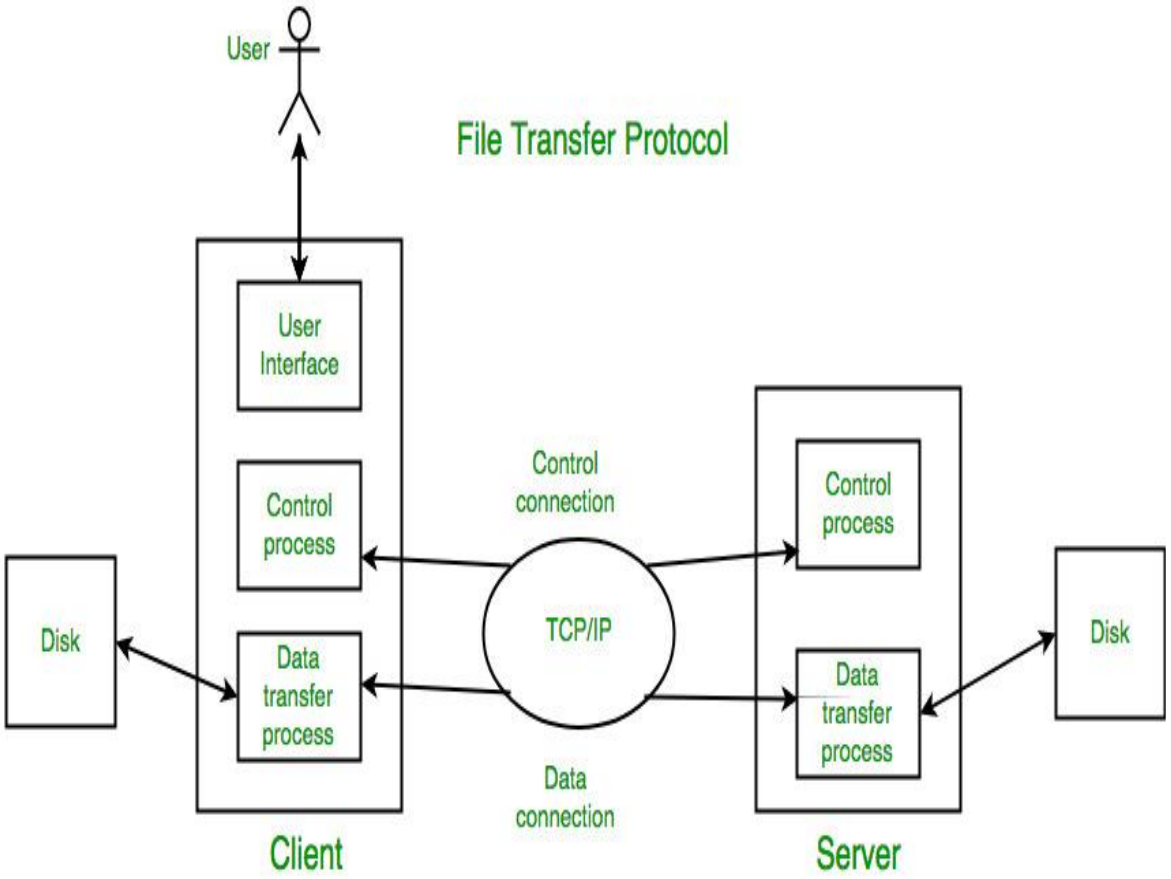
FILE TRANSFER PROTOCOL (FTP)

File Transfer Protocol (FTP) is an application layer protocol that moves files between local and remote file systems. It runs on top of TCP, like HTTP. To transfer a file, 2 TCP connections are used by FTP in parallel: control connection and data connection.

Why FTP?

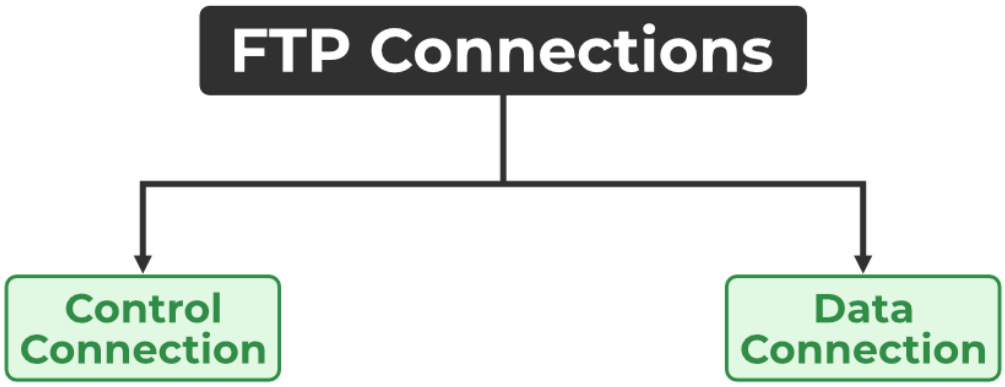
FTP is a standard communication protocol. There are various other protocols like HTTP which are used to transfer files between computers, but they lack clarity and focus as compared to FTP. Moreover, the systems involved in connection are heterogeneous systems, i.e. they differ in operating systems, directories, structures, character sets, etc the FTP shields the user from these differences and transfers data efficiently and reliably. FTP can transfer ASCII, EBCDIC, or image files. The ASCII is the default file share format, in this, each character is encoded by NVT ASCII. In ASCII or EBCDIC the destination must be ready to accept files in this mode. The image file format is the default format for transforming binary files.

Mechanism of File Transfer Protocol



Types of Connection in FTP

1. Control Connection
2. Data Connection



Types of Connection in FTP

1. Control Connection: For sending control information like user identification, password, commands to change the remote directory, commands to retrieve and store files, etc., FTP makes use of a control connection. The control connection is initiated on port number 21.
2. Data connection: For sending the actual file, FTP makes use of a data connection. A data connection is initiated on port number 20.
FTP sends the control information out-of-band as it uses a separate control connection. Some protocols send their request and response header lines and the data in the same TCP connection. For this reason, they are said to send their control information in-band. HTTP and SMTP are such examples.

FTP Session

When an FTP session is started between a client and a server, the client initiates a control TCP connection with the server side. The client sends control information over this. When the server receives this, it initiates a data connection to the client side. Only one file can be sent over one data connection. But the control connection remains active throughout the user session. As we know HTTP is stateless i.e. it does not have to keep track of any user state. But FTP needs to maintain a state about its user throughout the session.

FTP Clients

FTP works on a client-server model. The FTP client is a program that runs on the user's computer to enable the user to talk to and get files from remote computers. It is a set of commands that establishes the connection between two hosts, helps to transfer the files, and then closes the connection.
Some of the commands are:

get the filename(retrieve the file from the server)

get the filename(retrieve multiple files from the server)

ls(list files available in the current directory of the server)

There are also built-in FTP programs, which makes it easier to transfer files and it does not require remembering the commands.

FTP Data Structures

FTP allows three types of data structures :

1. **File Structure:** In file structure, there is no internal structure and the file is considered to be a continuous sequence of data bytes.
2. **Record Structure:** In record structure, the file is made up of sequential records.
3. **Page Structure:** In page structure, the file is made up of independent indexed pages.

FTP Commands

Some of the FTP commands are:

- USER – This command sends the user identification to the server.
- PASS – This command sends the user password to the server.
- CWD – This command allows the user to work with a different directory or dataset for file storage or retrieval without altering his login or accounting information.
- RMD – This command causes the directory specified in the path name to be removed as a directory.
- MKD – This command causes the directory specified in the pathname to be created as a directory.
- PWD – This command causes the name of the current working directory to be returned in the reply.
- RETR – This command causes the remote host to initiate a data connection and send the requested file over the data connection.
- STOR – This command causes the storage of a file in the current directory of the remote host.
- LIST – Sends a request to display the list of all the files present in the directory.
- ABOR – This command tells the server to abort the previous FTP service command and any associated transfer of data.
- QUIT – This command terminates a USER and if file transfer is not in progress, the server closes the control connection.

FTP Replies

Some of the FTP replies are :

- 200 – Command okay.
- 530 – Not logged in.
- 331 – User name okay, need a password.
- 225 – Data connection open; no transfer in progress.
- 221 – Service closing control connection.
- 551 – Requested action aborted: page type unknown.
- 502 – Command not implemented.
- 503 – Bad sequence of commands.
- 504 – Command not implemented for that parameter.

Characteristics of FTP

1. FTP uses TCP as a transport layer protocol.
2. It is good for simple file transfers, such as during boot time.
3. Errors in the transmission (lost packets, checksum errors) must be handled by the TFTP server.
4. It uses only one connection through well-known port 69.
5. TFTP uses a simple lock-step protocol (each data packet needs to be acknowledged). Thus the throughput is limited.

Advantages of FTP

- Speed is one of the advantages of FTP(File Transfer Protocol).

- File sharing also comes in the category of advantages of FTP in this between two machines files can be shared on the network.
- Efficiency is more in FTP.

Disadvantages of FTP

- File size limit is the drawback of FTP only 2 GB size files can be transferred.
- Multiple receivers are not supported by the FTP.
- FTP does not encrypt the data this is one of the biggest drawbacks of FTP.
- FTP is unsecured we use login IDs and passwords making it secure but they can be attacked by hackers.

AUDIO AND VIDEO

The new normal, where everything is happening online whether school, college, or office. People are meeting online, many conferences are held online, do you know what is common between them? Well the mode they all are meeting is through video conference and audio conference, without these technologies we can't think meeting in a group without any physical contact. So, let's understand audio conference and then video conference.

Audio conference

Also called Teleconference, means meeting a group or party via a phone call instead of meeting them physically. The person who starts the audio conference called the calling party and the people who are attending the calls are known as participants. So, in the audio conference at the same time, one can talk to many people. The benefit of using an audio conference as it is less expensive than the video conference because it needs very fewer devices but at the same time it lacks some richness of video conference.

Technology requirement of the audio conference:

1. Phone/Computer with access to internet connection
2. Speakers
3. Microphone

Benefits of Audio Conference

If we compare audio conference to video conference then it is **less expensive**, **easy to set up and** very much **accessible** by all. But, yes it can not totally compare to video conference as it lacks the richness of video conference. In a video conference, one can see each other as well as listen to one another, and this reduced miscommunication, but the audio conference is not the case.

The audio conference has also saved a lot of time and expense as now people do not have to travel to meet and discuss anything rather they can host the audio conference call. Also, at the same time, you can interact with as many people as you can.

Demerits of Audio Conference

Apart from the advantages, there are some limitations like the first one is the **communication is only verbal**. Studies show that effective communication is non-verbal communication. Also, the audio conference call **cannot keep you focused** in the meeting, as people cannot see images of each other so many times people who don't pay attention to the speaker and do his/her work like cooking or any either chores. The third demerit is the **poor quality of the audio call**, since the communication is telephonic then there are many times poor network reduced the quality of audio which affects the communication. Also, the **background noise** affects the quality of voice.

Video Conference

Basically, video conference enables the user to connect each other from different locations by both video and audio transmission at the same time, using telecommunication technology. Or we can say that it is a type of online meeting where people meet live on the audio-visual call. Video conference is also called video teleconference. It is more effective and efficient than the audio conference calls, as we can see the facial expression of the people which help us to connect them more easily rather than voice calls where only we can hear a voice. But, there are some problems, like setting video conference needs devices and all.

Technology requirement of video conference:

1. Computer with access to internet connection
2. Browser
3. Speakers
4. Microphone(for both audio and video conference)
5. Web camera (for only video conference)

Factors affecting Video Quality

If we talk about the factors which can affect video quality is firstly the speed of the internet. Poor internet connection or slow connection acts as a barrier because people aren't able to see each other, sometimes their video or audio got paused. Although with the advancement of new technology motion pictures can work in slower internet connection, but again people have to compromise with the video quality.

There are some minor factors too that can also affect the quality of video conferencing like the placement of the microphone. In video conferencing the audio plays an important role, without any audio one cannot decode the message of others. Thus, the placement and quality of the microphone also matter. At the same time, the placement of meeting, one should sit in the light before attending the video conference as in darkness your face will not visible to others. Also, during the presentation, the size of fonts and images matters a lot. If the font size of the presentation is too small or images are blurred then no one can read your presentation or understand your images.

Usefulness of Videoconferencing

The new normal already taught us a lesson about the importance of digital literacy. Nowadays, whether an office meeting or school all are happening from home, and they all are using video conferencing to meet each other. Video Conferencing also used to host live events and seminars. Now let's look at in detail :

1. **Distance Learning**: Even before the Covid situation, many people are using video conferencing in the education field. This enables to set up a classroom-type environment where teacher and student can both interact with each other. However, there are some limitations, and one is the lack of human touch.
2. **Business meetings**: Again this is very popular, people meet and do their work together over a distance.
3. **Telecommuting/Home Office**: No one has ever thought that people can work from home and the only thing which made it possible is the advancement of Information technology which includes video conference.
4. **Legal environment:** The use of video conferencing becomes a more frequent practice in the courtroom. It helps to testify witnesses to appear in court without physically travel to the courtroom.
5. **Telemedicine**: There are many times a patient needs medical advice but he/she is not able to visit a doctor physically, then video conferencing comes into the picture where staying at home one can take medical advice from the doctor or specialist.

So far we looked at both voice conference calls and video conference calls. We see both have some merits and demerits at the same time. Audio conference call is flexible, requires less expanse compared to video call and very easy to set up. At the same time video conference has a good quality of communication compares to audio because here one can see as well as hear each other. In a Video conference call, one can set up an environment like an office or a classroom which is not possible in audio conference calls. Thus, both have some goods and bad and this is up to us what we need